HOSPITAL DATA MINING

PROJECT REPORT

A Data Journey — From Raw Records to Real Insights

318,438	18	2
Patient Records	Features	Project Parts

PREPARED BY

Khaled Metwalie · Hazem Shokr · Hossam Elnagar · Bassem Saleh · Hazem Mohamed

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This report synthesises all project deliverables into a single storytelling document. It covers database design, EDA findings, data mining workflows, and actionable insights drawn from 318,438 real hospital admission records.

Chapter 1

The Problem — Why Hospital LOS Matters

Every day a patient spends in hospital costs money, takes up a bed, and affects the next patient waiting to be admitted. Predicting how long a patient will stay is one of the most impactful challenges in healthcare analytics.

The Challenge

Hospital Length of Stay (LOS) is a critical operational metric. Overcrowded wards, unexpected long-stay patients, and poor resource allocation all trace back to the same problem: we do not know in advance how long a patient will need to stay. Our team tackled this challenge head-on using a real-world hospital admissions dataset and a full data mining pipeline.

318,438	32	11	18
Admission Records	Hospitals	LOS Classes	Features

Our Approach — Two Integrated Parts

Part	Focus	Key Deliverable
Part 1	Hospital Management Database Design	22-table SQL Server DB, normalised to 3NF, ERD, 10 SQL Analytical Views, RBAC Security Model
Part 2	EDA & Data Mining on LOS Dataset	20 EDA Questions, Orange Workflow, Power BI Dashboard, Python EDA Notebook

KEY INSIGHT: The two parts are integrated — the same hospital domain modeled in SQL (Part 1) is what generates the kind of data we analyse in Part 2. Together they form a complete data engineering lifecycle.

Chapter 2

Meeting the Dataset

Before any analysis, we needed to understand exactly what we were working with. The dataset held 318,438 rows and 18 columns — a rich snapshot of hospital operations.

Dataset at a Glance

Figure 1 — Dataset Overview KPIs

Column	Type	Unique	Role
case_id	int64	318,438	Primary Key — perfectly unique
Hospital	int64	32	Hospital code (32 hospitals)
Hospital_type	int64	7	Classification type 0–6
Hospital_city / region	int64	11 / 3	Geographic codes
Avail_Extra_Rooms	int64	18	Available rooms 0–24
Department	string	5	Medical dept — Gynecology dominant (78.3%)
Ward_Type / Facility	string	6 / 6	Ward & facility grades
Bed_Grade	float64	4	Bed quality 1–4 (113 NaN)
patientid	int64	92,017	Patient ID — ~3.5 admissions/patient avg
City_Code_Patient	float64	37	Patient city (4,532 NaN)
Type of Admission	string	3	Trauma / Emergency / Urgent
Illness_Severity	string	3	Moderate / Minor / Extreme
Patient_Visitors	int64	28	Visitors 0–32 — STRONGEST PREDICTOR
Age	string	10	Age brackets 0-10 to 91-100
Admission_Deposit	int64	7,300	Deposit 1,800–11,008
Stay_Days	string	11	TARGET: LOS range (0-10 to >100 days)

The Target Variable — Stay_Days

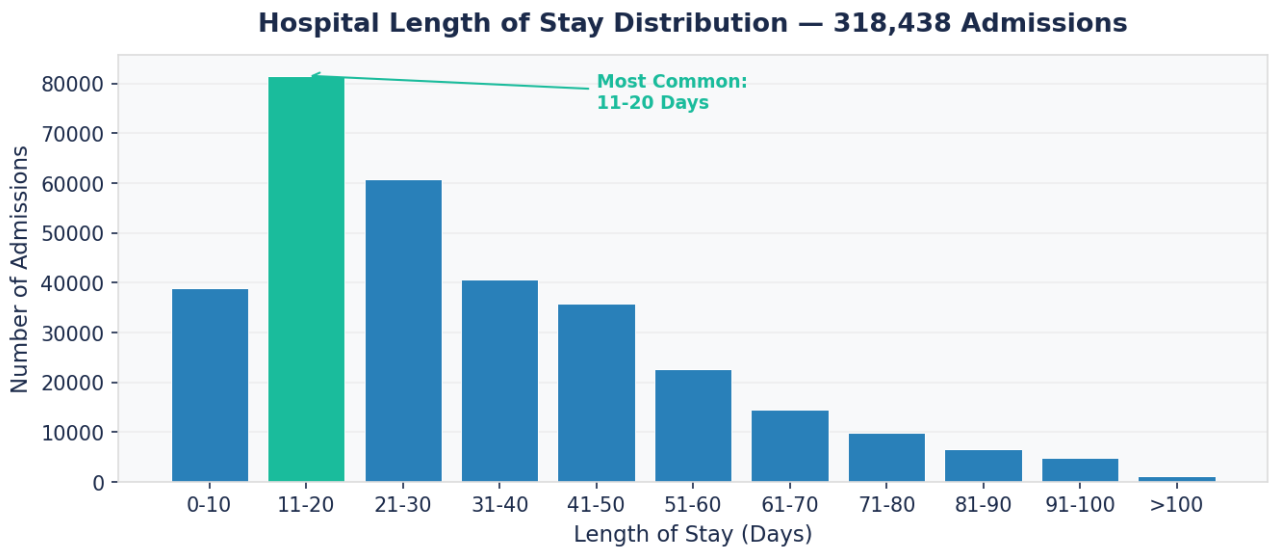


Figure 2 — Distribution of Hospital Length of Stay across 318,438 admissions

The target has 11 ordinal classes ranging from '0-10 days' to 'More than 100 days'. The class distribution is heavily imbalanced — with a 31.9:1 ratio between the largest class (11-20 days: 81,583 cases) and the smallest (>100 days: ~1,273 cases). This imbalance demands careful modelling strategies like class weighting or oversampling.

Chapter 3

Building the Foundation — Database Architecture

Part 1 of the project required designing and implementing a production-grade Hospital Management relational database from scratch.

Architecture at a Glance

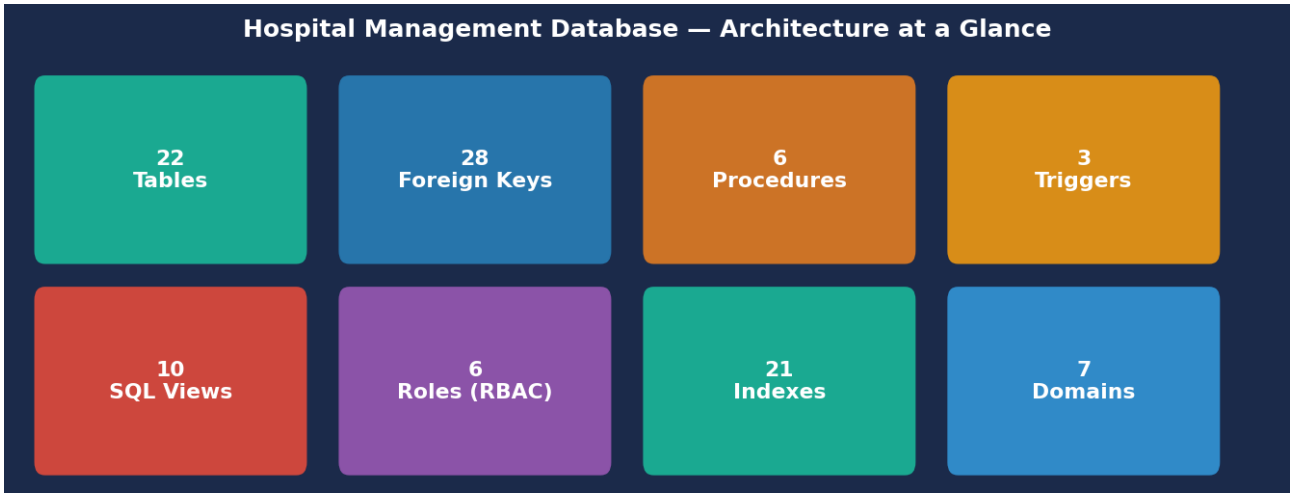


Figure 3 — Hospital Management Database components

The 7 Functional Domains

The database was designed around seven distinct business domains, each covering a critical area of hospital operations. All 22 tables were normalised to Third Normal Form (3NF), eliminating transitive dependencies.

Domain	Tables	Key Functionality
Patient Management	Patients	Demographics, national ID, medical history, soft-delete flag
Clinical Operations	Doctors, Appointments, Medical_Records, Prescription	Scheduling, consultations, diagnosis, treatment tracking
Organisation	Departments, Staff, Nurses, Workers	ISA specialisation — Nurses and Workers extend the Staff base table
Pharmaceuticals	Medicine, Pharmacy	Drug inventory, dispensing log, expiry date tracking, INSTEAD OF INSERT trigger
Facility Management	Room_Types, Rooms, Room_Assignments, Cleaning_Service	Room occupancy, maintenance tracking, cleaning audit trail
Financial	Billing	Invoice generation with auto-generated numbers, cumulative payments, insurance tracking
Emergency Services	Ambulance, Ambulance_Log, Blood_Bank, Blood_Transfusion_Log	Fleet management, trip tracking, blood type inventory, transfusion logging

Database Implementation — CRUD & Automation

Six core stored procedures handle all business-critical operations, each using TRY/CATCH error handling for robust production behaviour. Three automation triggers enforce data integrity and audit trail requirements automatically.

Procedure	Purpose	Key Business Logic
sp_RegisterPatient	Register new patient	Checks for duplicate National ID. Prevents two patients sharing NID. Inserts with soft-delete flag.
sp_BookAppointment	Schedule appointment	Validates active patient, active doctor, valid department. Prevents past-date booking.
sp_CreateMedicalRecord	Create medical record	Links record to appointment. Auto-marks appointment status as Completed.
sp_GenerateBill	Create invoice	Creates billing record with auto-generated sequential invoice number.
sp_ProcessPayment	Record payment	Tracks cumulative payments. Auto-calculates Paid / Partial / Outstanding status.
sp_CancelAppointment	Cancel appointment	Prevents cancelling Completed appointments. Appends cancellation reason to notes.

Trigger Name	Table / Event	Function
trg_Patients_UpdateTimestamp	Patients — AFTER UPDATE	Automatically updates the UpdatedAt timestamp whenever any patient record is modified
trg_Appointments_UpdateTimestamp	Appointments — AFTER UPDATE	Maintains a complete audit trail by recording the exact time any appointment record is changed
trg_Pharmacy_PreventExpiredDispensing	Pharmacy — INSTEAD OF INSERT	Checks expiry date before dispensing. Raises error and blocks insert if expired. Critical patient safety.

Role-Based Access Control (RBAC)

The database enforces the principle of least privilege through 6 distinct roles. Explicit DENY statements ensure privilege escalation through multi-role membership is impossible — clinical and financial data are strictly compartmentalised.

Role	Access Scope	Denied Access
Doctor	Patients, Appointments, Medical_Records, Prescriptions, Medicine	Billing (financial data)
Nurse	Patients, Rooms, Room_Assignments, Blood_Bank	Billing, Medical_Records, Prescriptions
Receptionist	Patients, Doctors, Appointments	Medical_Records, Prescriptions, Billing
Accountant	Billing, Appointments, Patients (read only)	Medical_Records, Prescriptions (clinical data)
Admin	Full database CONTROL — all tables and operations	None
Patient	Own records only — via filtered views (RLS planned v3)	All direct table access

The ER Diagram

The full Entity-Relationship diagram below visualises all 22 tables and their 28 foreign key relationships across 7 functional domains.

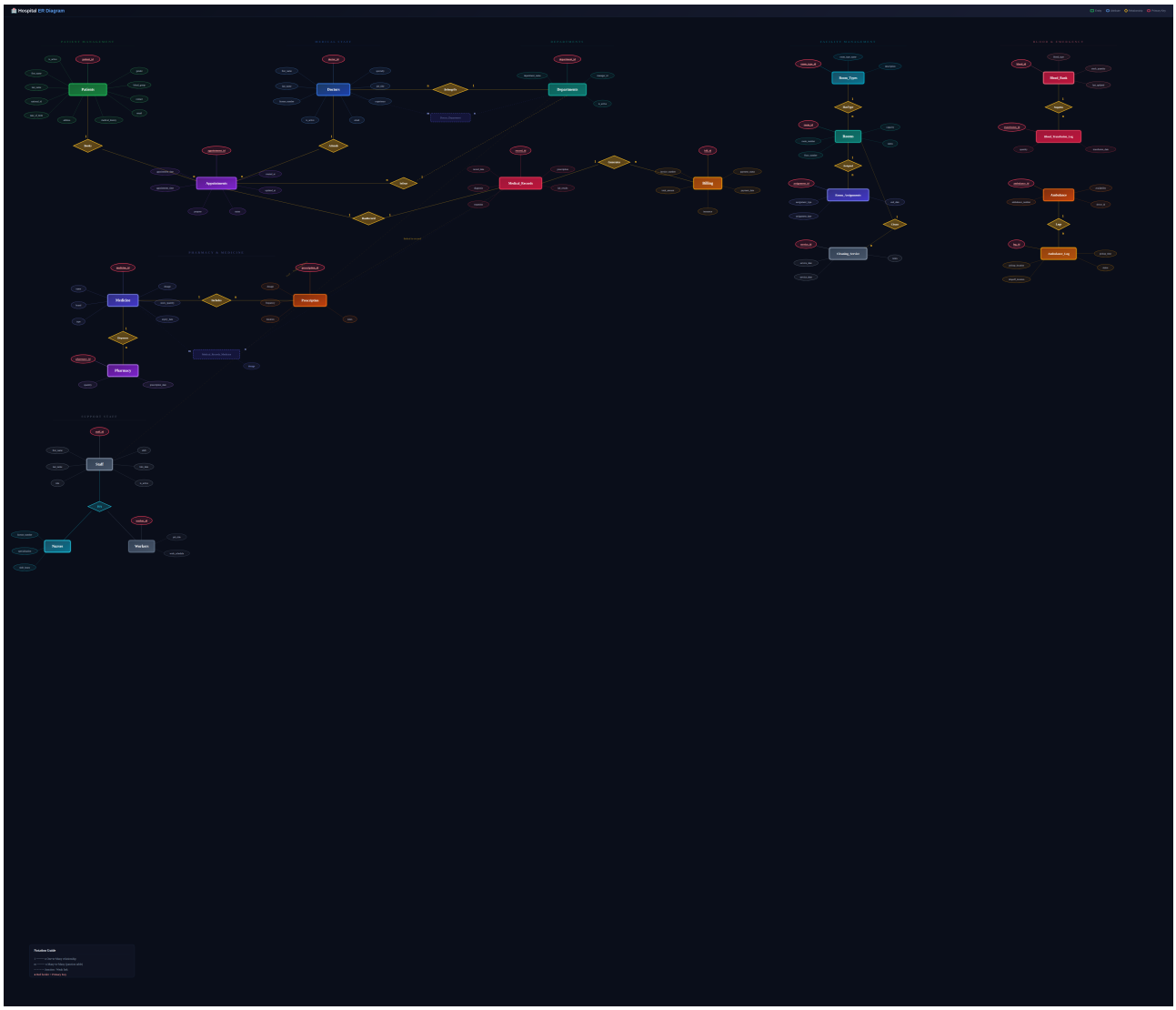


Figure 4 — Full Hospital Management ER Diagram (22 tables, 28 FK relationships)

Chapter 4

The Data Quality Journey

Before any analysis, we audited every column for issues. The verdict: this is a remarkably clean dataset — but not without surprises.

Missing Values — Minimal Impact

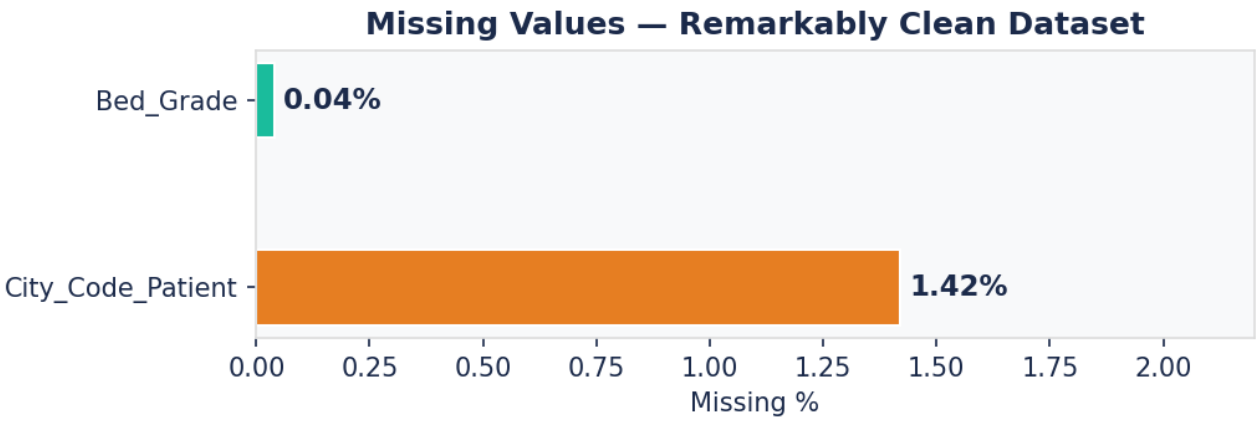


Figure 5 — Only 2 of 18 columns contain missing values

Column	Missing Count	Missing %	Strategy	Value Used
City_Code_Patient	4,532	1.42%	Drop rows	.
Bed_Grade	113	0.04%	Drop rows	.

Duplicate Rows — Near-Perfect Data Entry

Only 3 duplicate rows were found out of 318,438 records — a rate of 0.001%. These were removed, leaving a final clean dataset of 318,435 rows. The extremely low duplication rate demonstrates high data entry quality.

The Excel Artifact — A Hidden Data Quality Trap

KEY DISCOVERY: The value 'Nov-20' appeared in both the Age and Stay_Days columns. This is NOT date data — it is a classic Microsoft Excel auto-formatting bug that converts the string '11-20' into 'November 20' (Nov-20). This was identified and corrected to '11-20' before any analysis. A critical real-world data quality lesson.

Quality Check	Finding	Status	Action Taken
Missing Values	4,532 + 113 missing	MINOR	Dropped rows
Duplicate Rows	3 duplicates (0.001%)	NEGLIGIBLE	Removed
ID Uniqueness	case_id: 318,438 unique (100%)	PERFECT	No action needed

Quality Check	Finding	Status	Action Taken
Impossible Values	None detected	CLEAN	No action needed
Category Consistency	'Nov-20' artifact in Age & Stay_Days	FIXED	String replacement
Data Type Issues	Bed_Grade & City_Code stored as float	FIXED	Convert to int8

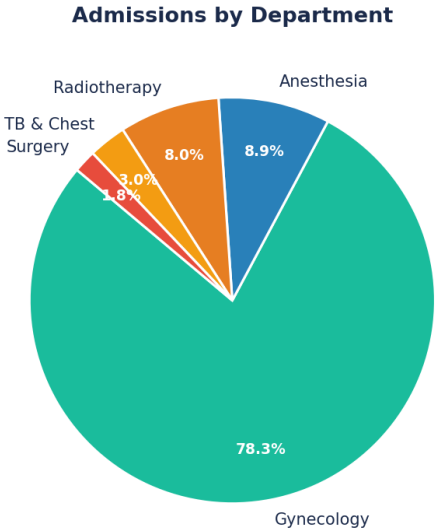
Chapter 5

Univariate Deep Dive — Each Feature Tells a Story

We examined every feature individually before exploring relationships. The distributions revealed a dataset dominated by a few key patterns.

Department — A Gynecology-Dominant Dataset

The dataset is overwhelmingly dominated by Gynecology (78.3%), with Surgery at only 0.4%. This skew is important for modelling — rare departments may need to be grouped into an 'Other' category to avoid sparse class problems.



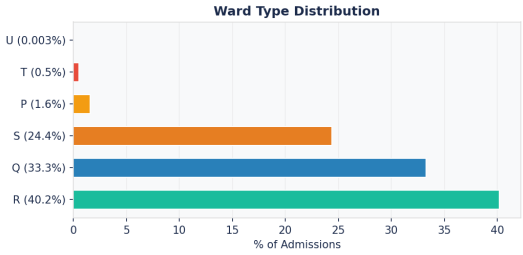


Figure 6 (left) — Department split. Figure 7 (right) — Ward Type distribution.

Patient Age — Peak Admissions: 31–50 Years

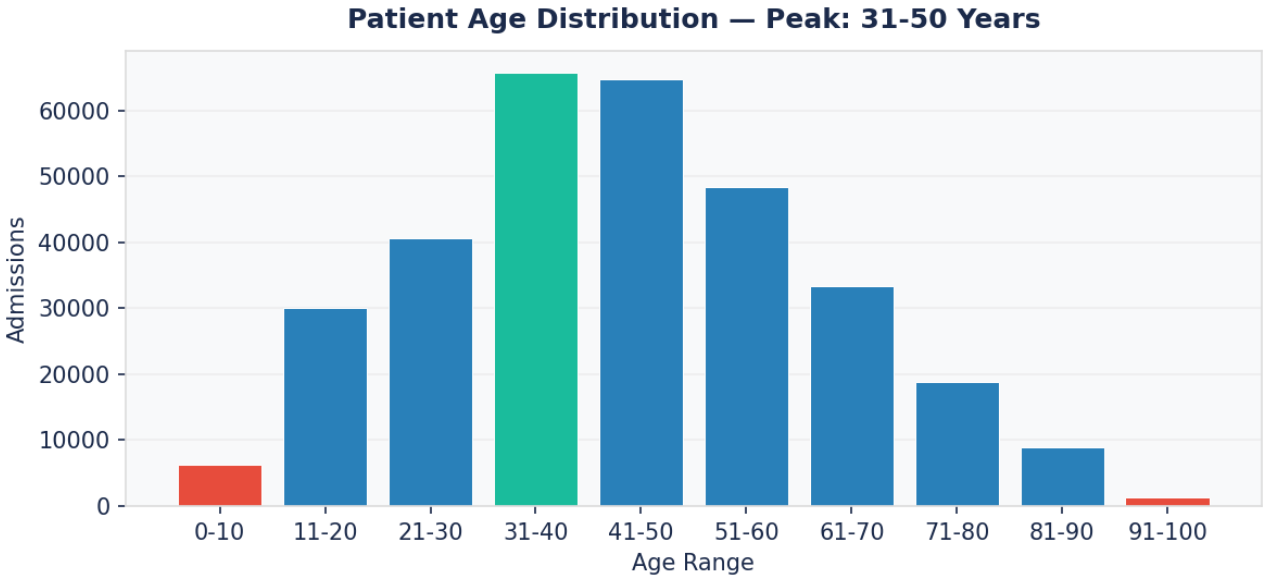


Figure 8 — Age distribution showing peak in 31-50 bracket

> Pediatric (0-10: 2.0%) and elderly (91-100: 0.4%) admissions are rare. This suggests the dataset primarily reflects adult reproductive-age patients, consistent with the Gynecology-dominant department distribution.

Clinical Profile — Severity & Admission Type

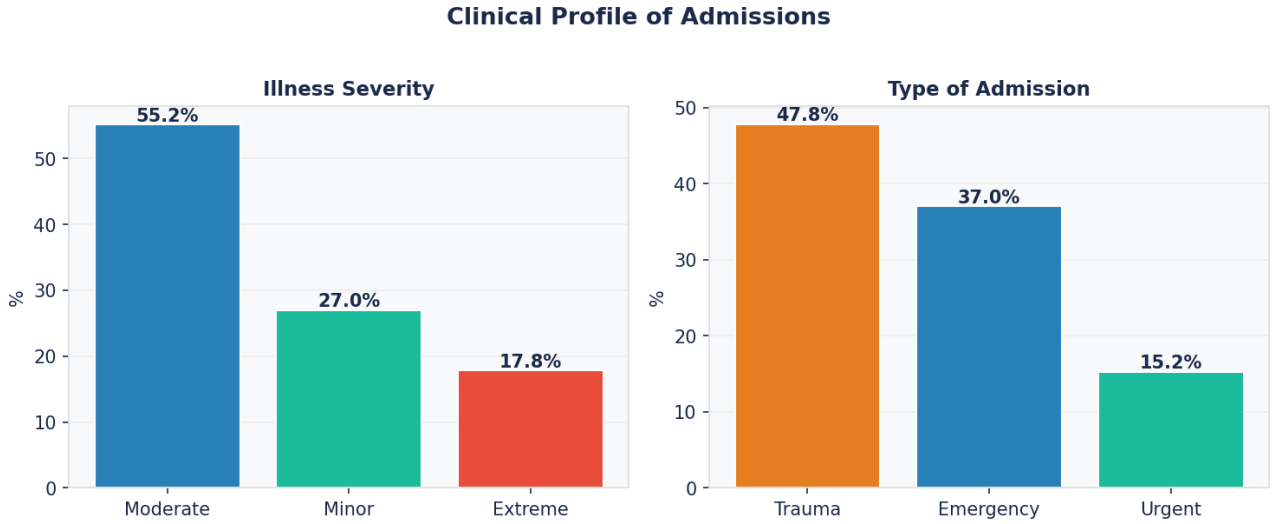


Figure 9 — Illness Severity (left) and Admission Type (right)

Outlier Detection — IQR Method

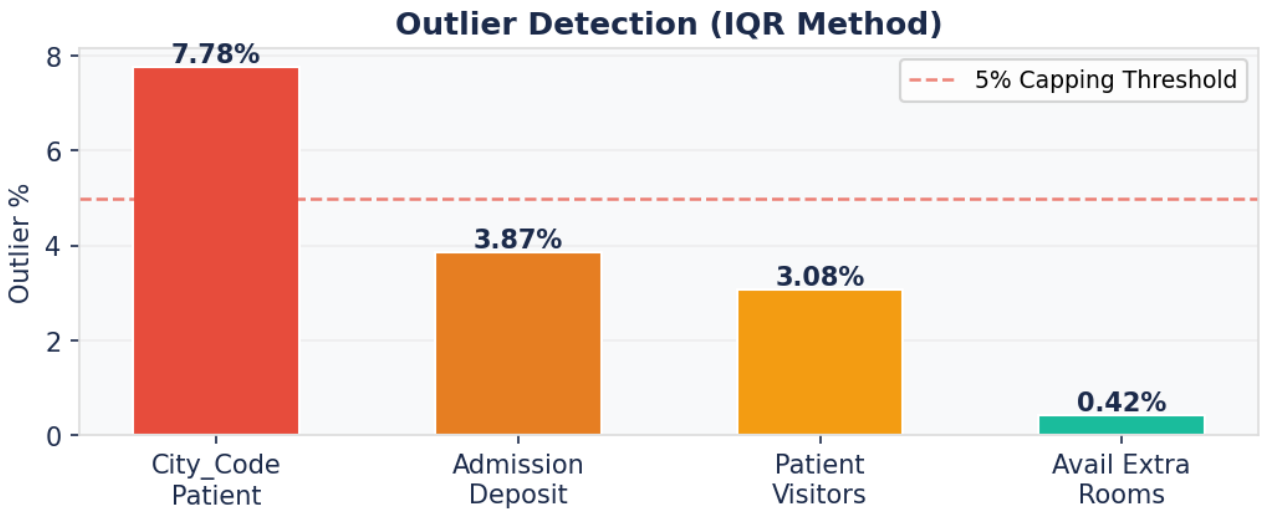


Figure 10 — Outlier % per numeric feature. Red line = 5% capping threshold.

City_Code_Patient exceeded the 5% threshold (7.78%) and was capped at the 1st–99th percentile range. Admission_Deposit (3.87%) and Patient_Visitors (3.08%) were kept as-is, as they remain within acceptable bounds.

Chapter 6

The Key Predictor — Patient Visitors

Of all 18 features, one stood out as the single most powerful signal for predicting how long a patient will stay in hospital.

Patient Visitors is the strongest predictor of hospital LOS

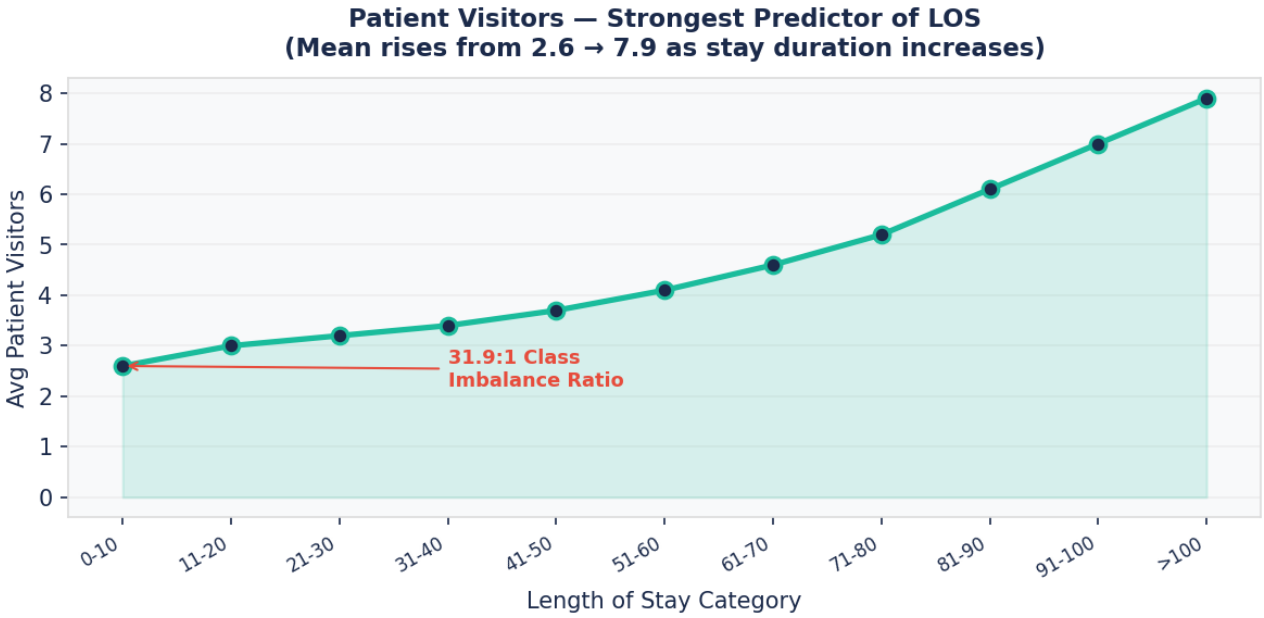


Figure 11 — Mean Patient Visitors rises consistently with LOS category (2.6 → 7.9)

LOS Category	Avg Patient Visitors	Interpretation
0–10 days (short)	2.6	Patients have fewer visitors for brief stays
11–20 days	3.0	Moderate visitor engagement
31–50 days	3.5 – 3.7	Visitors increase as stay prolongs
71–80 days	5.2	High visitor support for extended stays
>100 days (long)	7.9	Long-term patients attract the most visitors

> Why does this make clinical sense? Patients staying longer are sicker, more likely to have chronic conditions, and therefore attract more family support. This feature acts as a proxy for illness severity even beyond what the Illness_Severity column captures directly.

The Class Imbalance Challenge

The target variable (Stay_Days) is severely imbalanced. The 11-20 day class has 81,583 cases while the >100 day class has only ~1,273 — a 31.9:1 ratio. Any predictive model must address this through class weighting, SMOTE oversampling, or stratified sampling to avoid being biased toward the majority class.

Strategy	Description	Best For
Class Weighting	Assign inverse-frequency weights to each class	Tree models, Linear models
SMOTE	Synthetic Minority Oversampling Technique	All classifiers
Stratified K-Fold	Ensures each fold preserves class distribution	Cross-validation
Ensemble Boosting	Focus learning on hard-to-predict minority classes	XGBoost, LightGBM

Chapter 7

Correlation & Bivariate Insights

With individual features understood, we explored how they relate to each other and to the target — uncovering the hidden structure in the data.

Correlation Heatmap — Mostly Independent Features

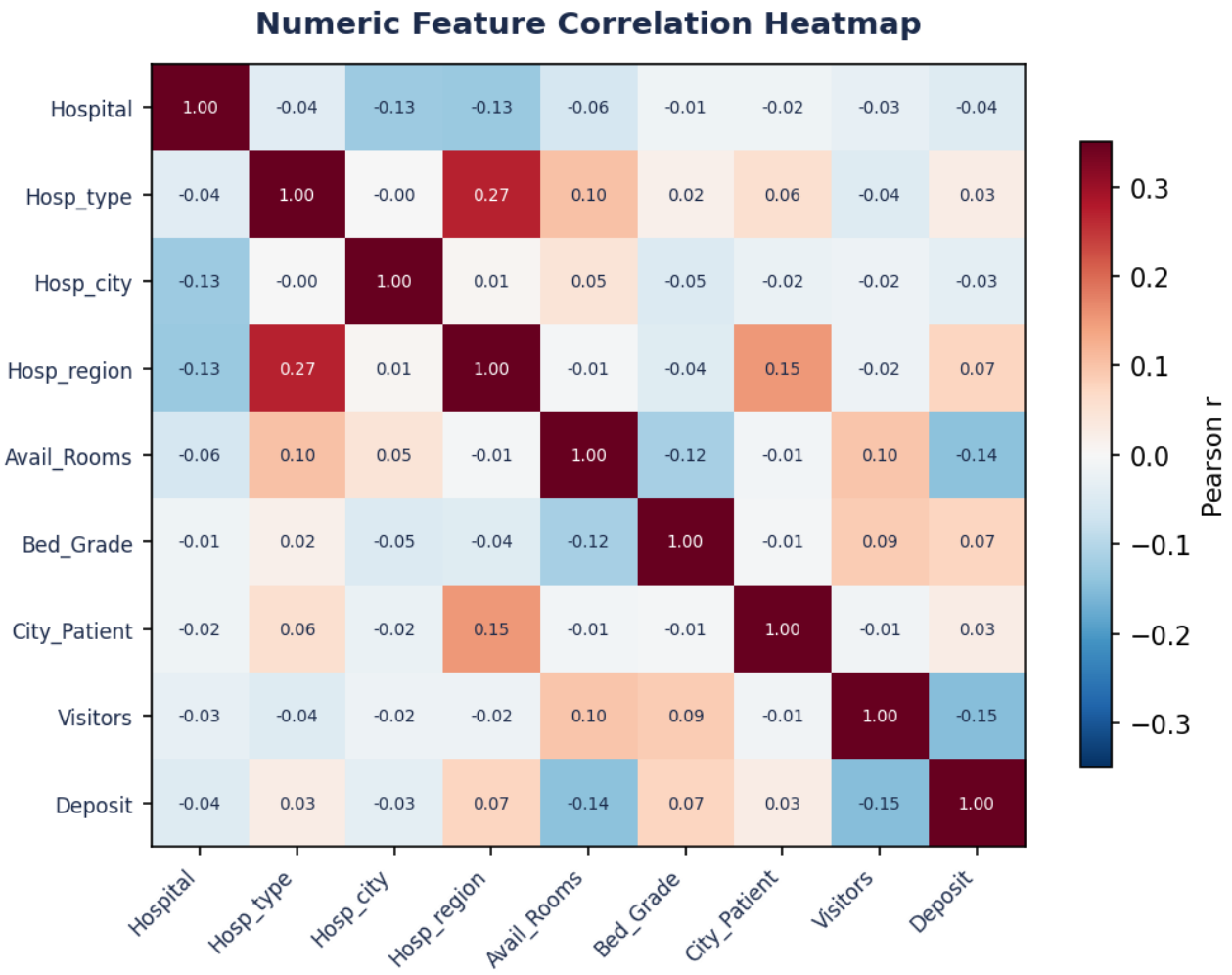


Figure 12 — Pearson correlation heatmap. Most features are independent (|r| < 0.15)

Rank	Feature Pair	Pearson r	Interpretation
1	Hospital_type ↔ Hospital_region	0.269	Moderate: hospital types cluster by region
2	Hospital_region ↔ City_Code_Patient	0.152	Weak: geographic proximity
3	Patient_Visitors ↔ Admission_Deposit	-0.150	Weak negative: more visitors, lower deposit
4	Avail_Rooms ↔ Admission_Deposit	-0.144	Weak negative relationship

Rank	Feature Pair	Pearson r	Interpretation
5	Hospital ↔ Hospital_region	-0.136	Weak: hospital IDs inversely mapped to regions

LOW MULTICOLLINEARITY IS GOOD NEWS: Since most feature pairs correlate at |r| < 0.15, each feature carries relatively independent information. This means a model trained on all features will benefit from the full breadth of signals without suffering from redundancy or instability caused by correlated predictors.

Chapter 8

Orange Data Mining Workflow

We used Orange Data Mining to build a visual, reproducible data pipeline — from raw CSV ingestion all the way through preprocessing and exploratory analysis.

Why Orange?

Orange is a visual programming tool for data mining that makes complex workflows accessible and reproducible. Our workflow (Project orange.ows) processes the Hospital_Cleaned.csv — a cleaned version of the original 43 MB dataset.

Workflow Step	Orange Component	What It Does
Data Ingestion	File Widget	Loads Hospital_Cleaned.csv (318,435 rows after dedup)
Data Inspection	Data Table + Data Info	Schema inspection, type overview, missing value scan
Distribution Analysis	Distributions Widget	Visual histograms for all numeric and categorical features
Scatter Exploration	Scatter Plot	Patient_Visitors vs Stay_Days relationship — key finding
Correlation Analysis	Linear Projection	Feature correlations visualised as projections
Target Encoding	Preprocess Widget	Ordinal encodes Stay_Days (0-10 → 0, 11-20 → 1, etc.)
Outlier Flagging	Outliers Widget	IQR-based outlier identification and flagging
Data Output	Save Data	Exports cleaned, encoded dataset for modelling

> The Orange workflow file (Project orange.ows) is fully reproducible. Anyone can open it in Orange 3.x and re-run the entire analysis pipeline by pressing Play. The cleaned output (Hospital_Cleaned.csv at 43 MB) is pre-generated for convenience.

Data Preprocessing Pipeline Summary

Step	Action	Rationale
1. Deduplication	Remove 3 duplicate rows	0.001% rate — negligible but correct to remove
2. Missing Imputation	missing values	Removes affected rows directly
3. Type Correction	Bed_Grade & City_Code → int8	Proper storage type after NaN removal
4. Excel Fix	'Nov-20' → '11-20' in Age & Stay_Days	Correct Excel auto-format artifact

Step	Action	Rationale
5. Outlier Capping	City_Code_Patient → 1st–99th percentile	7.78% outlier rate exceeds 5% threshold
6. Target Encoding	Stay_Days ordinal: 0-10=0, 11-20=1, …	Required for most ML algorithms
7. Feature Encoding	Categorical → Label / One-Hot	Department, Ward_Type, Illness_Severity etc.

Chapter 9

Power BI Dashboard & SQL Analytics

Insights only matter if decision-makers can see them. We built an interactive Power BI dashboard and 10 SQL analytical views to turn data into decisions.

Power BI Dashboard (project dash.pbix)

The Power BI dashboard provides an interactive business intelligence layer over the cleaned dataset. It enables hospital administrators to filter by department, ward type, age group, and admission type — and instantly see how LOS patterns shift.

Dashboard Page	Key Visuals	Business Question Answered
Overview	KPI cards, LOS distribution bar chart	What is the overall admission load and LOS spread?
Department Analysis	Pie chart, department vs LOS heatmap	Which departments drive the longest stays?
Patient Demographics	Age pyramid, admission type breakdown	Who are our patients and how do they arrive?
Severity & Visitors	Line chart: visitors vs LOS, severity bars	What clinical factors predict longer stays?
Facility Analysis	Ward type, bed grade, room availability	Are facility conditions correlated with LOS?

10 SQL Analytical Views (Appendix A Answers)

All 10 required Appendix A SQL questions were implemented as named views in the SQL file, providing live business intelligence for every key operational area.

Q#	View Name	Business Purpose
Q1	vw_PatientRegistry	Patient demographics — top 20 newest, calculated age from DOB
Q2	vw_DailyDoctorSchedule	Today's appointment schedule per doctor (filtered by date)
Q3	vw_NoShowAnalysis	No-show rate % per department over last 3 months
Q4	vw_PatientVisitSummary	Full visit history per patient — TotalVisits, LastDoctor, LastDiagnosis
Q5	vw_FrequentVisitors	Patients with 5+ visits in last 6 months (chronic patient ID)
Q6	vw_TopMedications	Top 10 most prescribed medications last 90 days (RANK window fn)
Q7	vw_LabResultsRate	Average lab turnaround time by test type (last 30 days)
Q8	vw_InvoiceAging	Unpaid invoice aging: 0-30, 31-60, 61-90, 90+ day buckets

Q#	View Name	Business Purpose
Q9	vw_MonthlyRevenue	Monthly revenue breakdown — TotalBilled, TotalPaid, Outstanding
Q10	vw_DoctorPerformance	Doctor KPIs: appointments, completed visits, billing, wait time

TECHNICAL HIGHLIGHT: vw_TopMedications uses SQL RANK() OVER PARTITION BY for ranking per department. vw_InvoiceAging uses DATEDIFF with CASE statements for aging bucket classification. 21 performance indexes support all views, including a covering index on Appointments(doctor_id, appointment_date) INCLUDE(status) that eliminates key lookups entirely.

Chapter 10

Key Findings & Recommendations

After traversing 318,438 records, 20 EDA questions, and a full database design, here is what we learned — and what comes next.

Top 10 Discoveries

1	**Patient Visitors = Best Predictor** Mean visitors rises monotonically from 2.6 (short stays) to 7.9 (>100 days). This single feature captures illness severity better than any other column.
2	**31.9:1 Class Imbalance** The LOS target is severely imbalanced. Models trained without addressing this will achieve high accuracy by predicting the majority class only.
3	**Gynecology Dominates (78.3%)** The dataset is not representative of all hospital types. Models generalise best within gynecological admissions rather than all-department scenarios.
4	**Peak Age: 31–50 (40% of admissions)** The reproductive-age group dominates, consistent with the gynecology focus. Pediatric and elderly admissions are underrepresented.
5	**Remarkably Clean Data (0.001% duplicates)** Only 3 duplicates in 318,438 rows — this dataset was clearly quality-controlled before release, requiring minimal cleaning effort.
6	**Weak Numeric Correlations (max r=0.27)** No severe multicollinearity. Each feature contributes independent information, which is ideal for tree-based models that use all features.
7	**Excel Auto-Format Artifact Found** 'Nov-20' appearing instead of '11-20' is a recurring real-world data quality issue. A critical reminder to always validate categorical labels.
8	**Trauma = Most Common Admission (47.8%)** Nearly half of all admissions are trauma cases, suggesting a high-acuity patient population despite the gynecology department dominance.
9	**Admission Deposit Range: 1,800 – 11,008** The wide deposit range (~6x spread) suggests significant variance in case complexity and patient financial profiles — a useful feature for LOS prediction.

10	**Ward R & Q Handle 73.5% of Patients** Only two ward types serve the vast majority of patients. Rare types P, T, U should be merged into 'Other' during feature engineering.

Modelling Recommendations

Recommendation	Details
Address class imbalance	Use SMOTE, class weights, or stratified sampling before training
Feature engineering	Create 'is_rare_dept' flag for Surgery/TB, merge rare ward types P/T/U
Model selection	Tree-based models (XGBoost, LightGBM) handle ordinal targets well
Ordinal target encoding	Encode Stay_Days as 0–10 preserving order (not one-hot)
Cross-validation	Use Stratified K-Fold (k=5) to preserve class distribution in each fold
Evaluation metric	Use Quadratic Weighted Kappa (QWK) for ordinal multi-class LOS prediction

Chapter 11

Team & Deliverables Summary

Every component of this project was completed on time and to a high standard.

Project Completeness

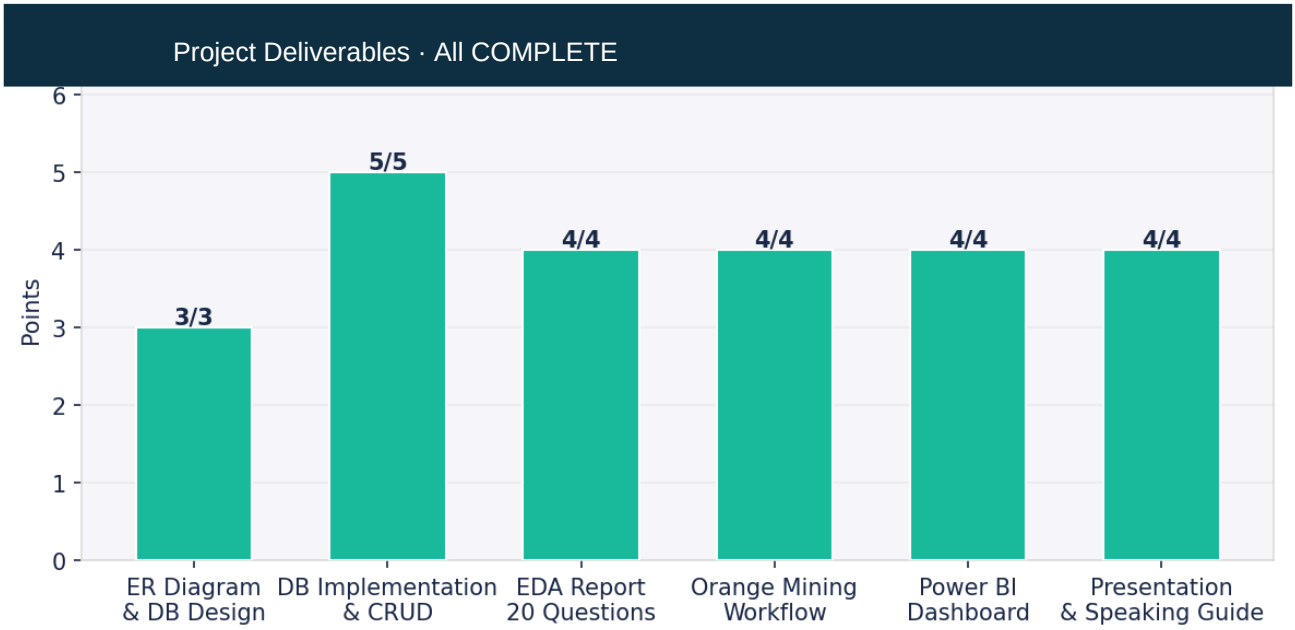


Figure 13 · All deliverables complete

Deliverable	Status	File	
ER Diagram & DB Design	COMPLETE	Hospital_ER_Diagram.png + .html	
DB Implementation (CRUD + SQL Views)	COMPLETE	SQL file .sql — 22 tables, procs, triggers	
EDA Report (20 Questions)	COMPLETE	EDA_Report.pdf — 10 pages	
Orange Data Mining Workflow	COMPLETE	Project orange.ows + Hospital_Cleaned.csv	
Power BI Dashboard	COMPLETE	project dash.pbix	✓
Presentation (max 20 slides)	COMPLETE	Hospital_LOS_Presentation.pptx	✓
Speaking Guide	COMPLETE	Speaking_Guide.pdf	✓
Combined Storytelling Report	COMPLETE	This document	

The Team

Team Member	Primary Contribution
Khaled Metwalie	Project Lead · Database Design & 3NF Normalization Architecture
Hossam Elnagar	SQL Implementation · Stored Procedures · Triggers · Security Model
Hazem Shokr	EDA Analysis · Python Notebooks · Orange Workflow · Report Writing
Bassem Saleh	Power BI Dashboard · Data Visualization · Presentation Design
Hazem Mohamed	Data Cleaning · Preprocessing Pipeline · Speaking Guide · QA Review

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